On the Proof of the Prime Number Theorem

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The Distribution of Prime Numbers

PNT

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The ratio $\pi(x)/(x/\log x)$ tends to 1 as x approaches infinity.

Introduction



In 1792

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as $x \to \infty$

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as $x \to \infty$

In 1798

$$\pi(x) \approx x/(\log x - A)$$

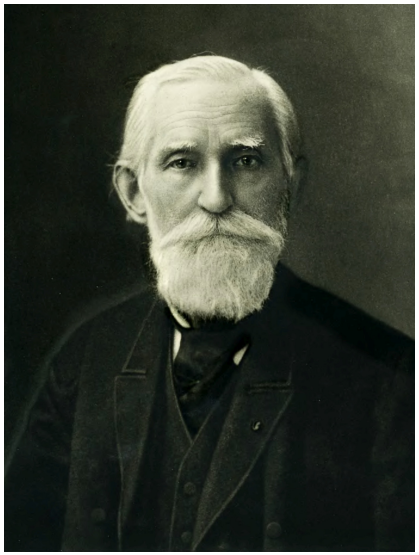
with constant

$$A \approx 1.08366.$$



Introduction

In 1850s



$$0.92\ \frac{x}{\log x} \leq \pi(x) \leq 1.11\ \frac{x}{\log x}$$

$$\lim_{x\to\infty} \frac{\pi(x)}{(x/\log x)} = 1$$

Introduction



In 1859

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s} \quad (Re(s) > 1)$$

$$\zeta(s) = \prod_{p} \left(1 - p^{-s}\right)^{-1}$$

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In 1896

$\zeta(s)$ has no zeros
on the line
$Re(s) = 1$



Proving the Prime Number Theorem

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The Riemann
Zeta Function

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Derivative Function

$$\Phi(s) := -\frac{d}{ds}\left(\sum_{p \in \mathscr{P}} \frac{1}{p^s}\right)$$

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Chebyshev's
ϑ**-Function**

$$\vartheta(x) = \sum_{p \leq x} \log p$$

$$(x > 1)$$

Lemma 1

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The Idea of the Proof

As $\prod_{p \in \mathscr{P}} (1 + 2z)$ converges, we bound the infinite product:

$$\prod_{p \in \mathscr{P}} \frac{1}{1 - \frac{1}{p^s}} \leq \prod_{p \in \mathscr{P}} \left(1 + \frac{2}{p^s}\right) \leq \infty$$

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$$\sum_{p \in \mathscr{P}} \left|\frac{2}{p^s}\right| = 2 \sum_{p \in \mathscr{P}} \left|\frac{1}{p^{Re(s)}}\right| \leq 2 \sum_{n=1}^{\infty} \frac{1}{n^{Re(s)}} < \infty \; , \; Re(s) > 1$$

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$$\sum_{n=1}^{\infty} \frac{1}{n^s} \leq \prod_{p \in \mathscr{P},\ p<N} \left(1 + \frac{1}{p^s} + \frac{1}{p^{2s}} + \dots\right) \overset{N \to \infty}{\Longrightarrow} \zeta(s) \leq \prod_{p \in \mathscr{P}} \frac{1}{1 - \frac{1}{p^s}}$$

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$$\zeta(s) - \frac{1}{s-1} = \sum_{n \geq 1} \underbrace{\int_n^{n+1} \left(\frac{1}{n^s} - \frac{1}{x^s} \right) dx}_{:= a_n(s)}$$

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The **Mean Value Inequality**, for $x \in [n, n+1]$

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Since $x \in [n, n+1]$,

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Thus

$$|a_n(s)| \leq \frac{|s|}{n^{Re(s)+1}} \int_n^{n+1} (x-n)dx = \frac{|s|}{n^{Re(s)+1}} \cdot \frac{1}{2}$$

Lemma 3

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Thus: $\displaystyle\prod_{n<p\leq 2n} p$ divides $\binom{2n}{n}$ which implies $\displaystyle\prod_{n<p\leq 2n} p \leq \binom{2n}{n} \leq 4^n$

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then, we get by monotonicity of ϑ

$$\vartheta(x) \leq \vartheta(2^{k_x+1}) \leq 2^{k_x} \cdot 2\log 4 \leq x \cdot 4\log 2$$

Lemma 4

Outside the pole at $s = 1$, the holomorphic function $\zeta(s)$ in $\{Re(s) > 0\} \setminus \{1\}$ never vanishes on $\{Re(s) = 1\}$:

$$\boxed{\zeta(1 + it) \neq 0, \quad \forall t \in \mathbb{R}^*}$$

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$$\frac{\zeta'(s)}{\zeta(s)} = \frac{d}{ds} \log\left(\zeta(s)\right) = \frac{d}{ds} \log\left(\prod_p \left(1 - \frac{1}{p^s}\right)^{-1}\right)$$

$$\frac{\zeta'(s)}{\zeta(s)} = \underbrace{\sum_{p \in \mathscr{P}} \frac{\log p}{p^s}}_{=\Phi(s)} + \underbrace{\sum_{p \in \mathscr{P}} \frac{\log p}{p^s(p^s - 1)}}_{=\mathscr{R}(s)}$$

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Contradiction.

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Thus, near s_0,

$$\frac{\zeta'(s)}{\zeta(s)} \approx \frac{k}{s - s_0}$$

Contradiction. Thus, on $Re(s) = 1$, $\zeta(s)$ cannot vanish there.

Lemma 5

The function

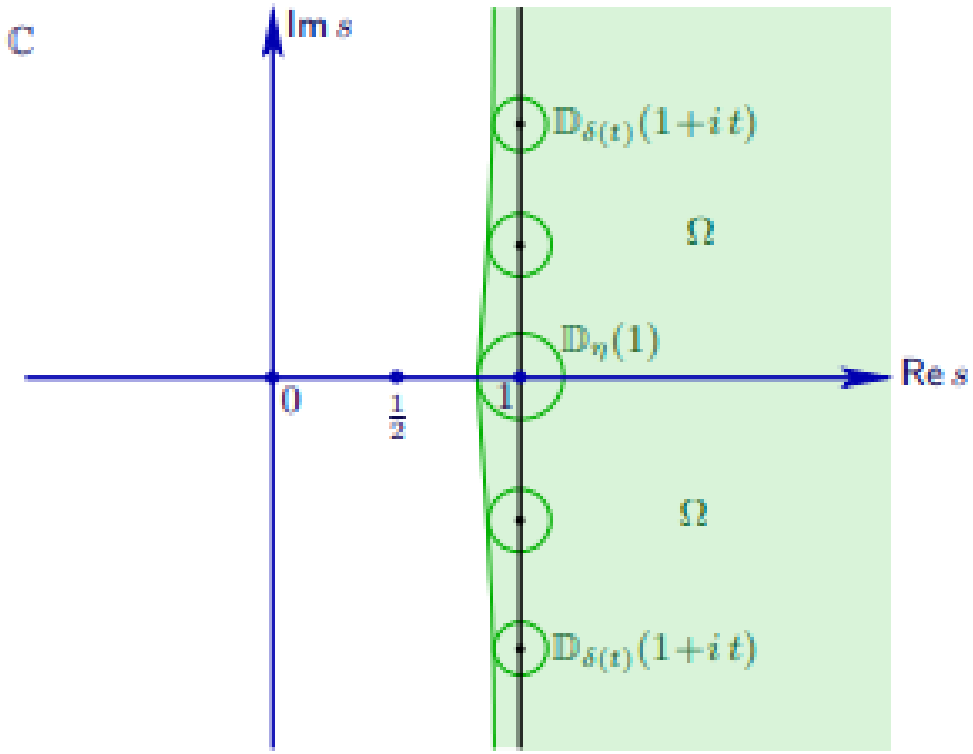
$$\Phi - \frac{1}{s-1} = -\frac{\zeta'(s)}{\zeta(s)} - \frac{1}{s-1} - \sum_{p \in \mathscr{P}} \frac{\log p}{p^s(p^s-1)}$$

is holomorphic in a certain open neighbourhood:

$$\boxed{\Omega \supset \{Re\, s \geq 1\}}$$

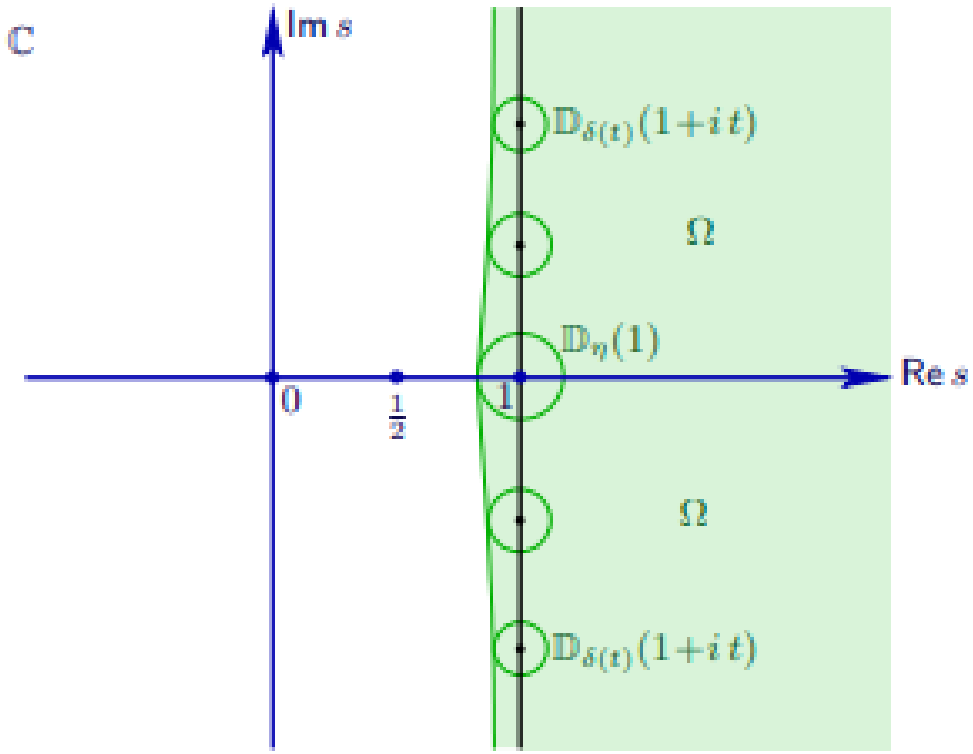
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$$\Omega = \{Re(s) > 1\} \cup D_{\eta}(1) \cup \bigcup_{t \in \mathbb{R}^*} D_{\delta(t)}(1+it)$$

$$\Omega \supset \{Re(s) \geq 1\}$$

Lemma 6

The integral:

$$\int_1^{\infty} \frac{\vartheta(x) - x}{x^2}\, dx = \lim_{\tau \to \infty} \int_1^{\tau} \frac{\vartheta(x) - x}{x^2}\, dx \quad \text{Converges}$$

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We have $$\Phi(s) = \sum_p \frac{\log p}{p^s} = \int_1^{\infty} \frac{1}{x^s}\, d\vartheta(x)$$

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We have $$\Phi(s) = \sum_p \frac{\log p}{p^s} = \int_1^{\infty} \frac{1}{x^s}\, d\vartheta(x)$$

Using integration by parts in the Riemann-Stieltjes integral

$$\Phi(s) = s \int_1^{\infty} \frac{\vartheta(x)}{x^{s+1}}\, dx$$

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$$g(0) = \int_0^\infty f(t)\,dt = \int_0^\infty \left(\vartheta(e^t)e^{-t} - 1\right)dt = \int_1^\infty \frac{\vartheta(x) - x}{x^2}\,dx$$

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confirming the existence of the integral.

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The Idea of the Proof

$\vartheta(x) \geq \lambda x$ for some $\lambda > 1$. Suppose there exists a sequence $(x_n)_{n=1}^{\infty}$ with $x_n \to \infty$ such that: $\vartheta(x_n) \geq \lambda x_n$.

Lemma 7

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Violates Lemma 6.

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Again, contradicting lemma 6.

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This gives the lower bound for $\pi(x)$:

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Rearranging:

$$\pi(x) \leq \frac{x}{(1-\varepsilon)\log x} + x^{1-\varepsilon} \to \pi(x) \leq \frac{x}{(1-\varepsilon)\log x} \to \pi(x) \leq \frac{x}{\log x}$$

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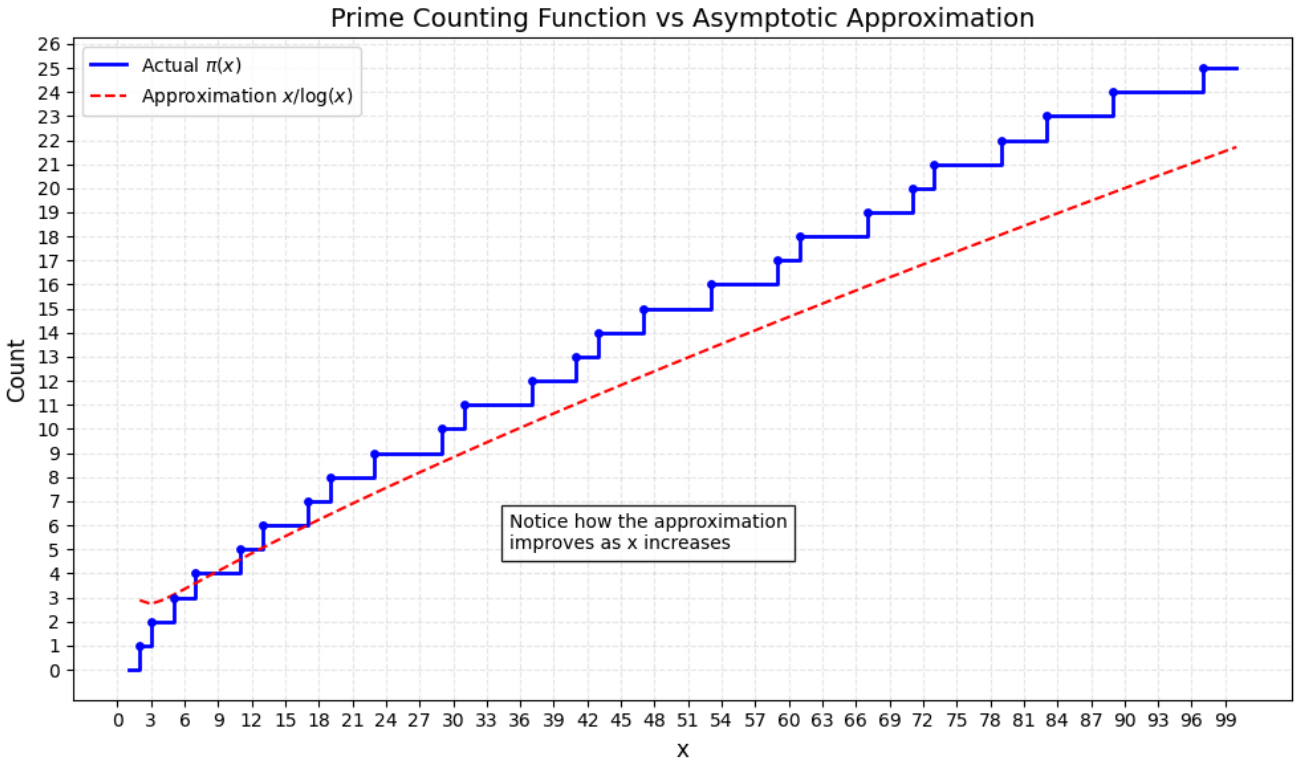
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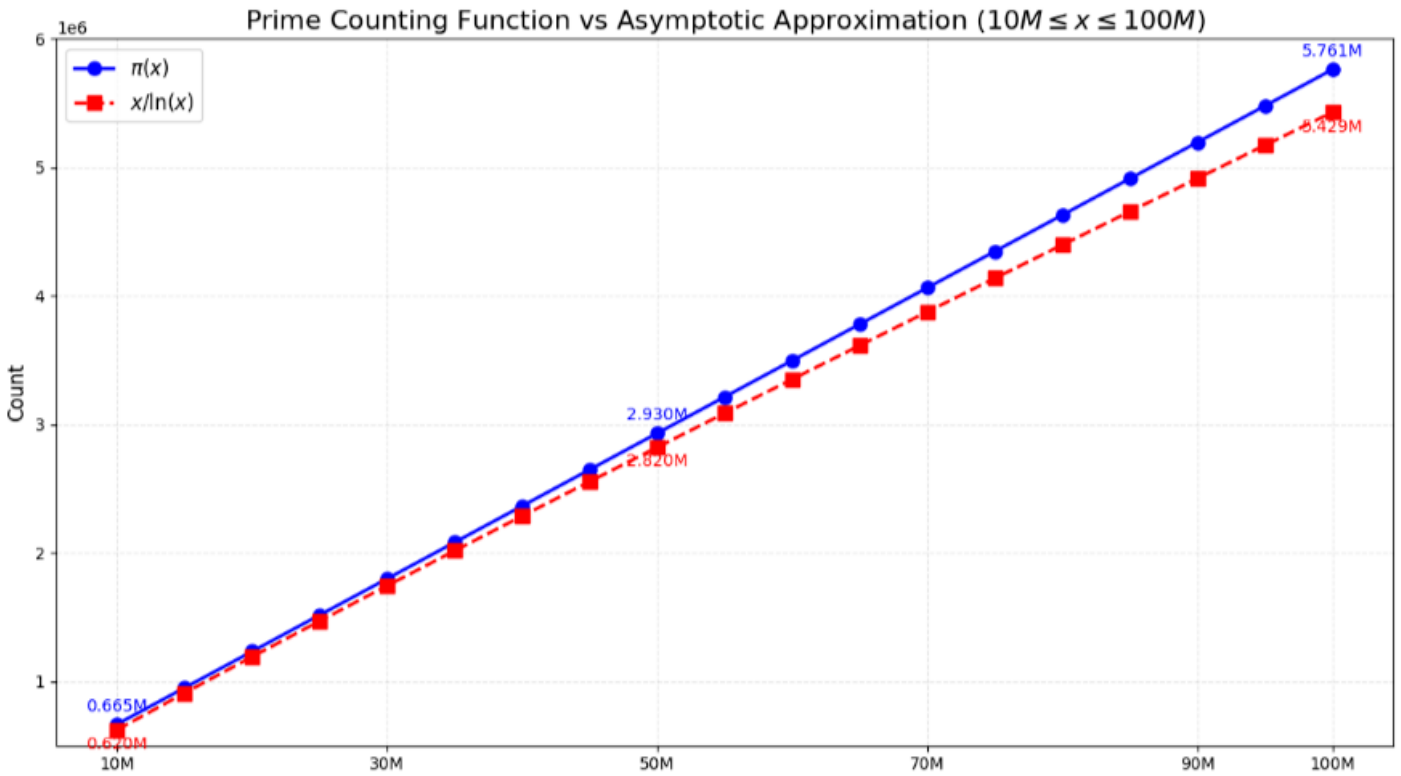
This completes the proof of the prime number theorem.

Final Thoughts

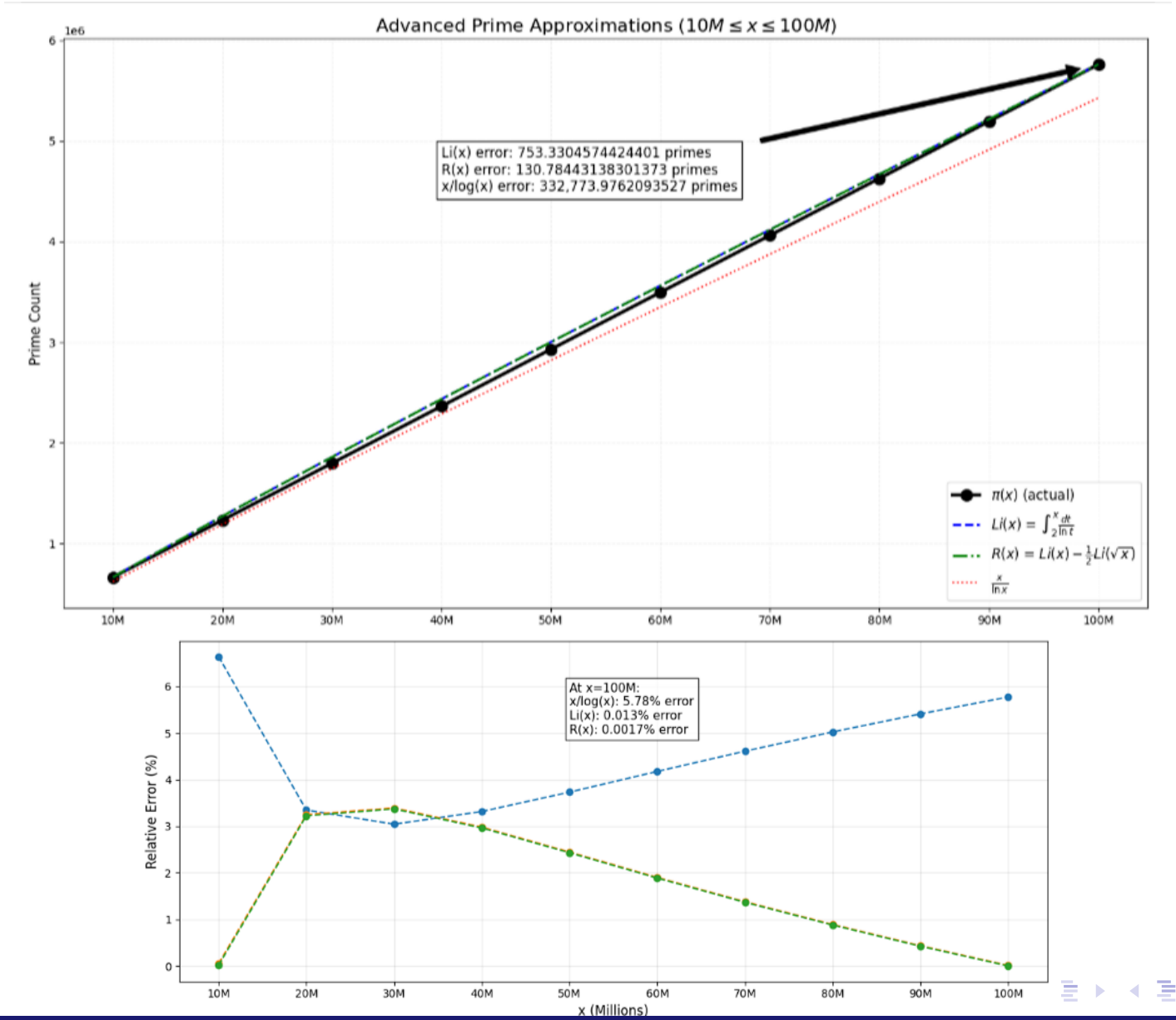
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Thank you!